Observer | Preserve the interface, add a recorder

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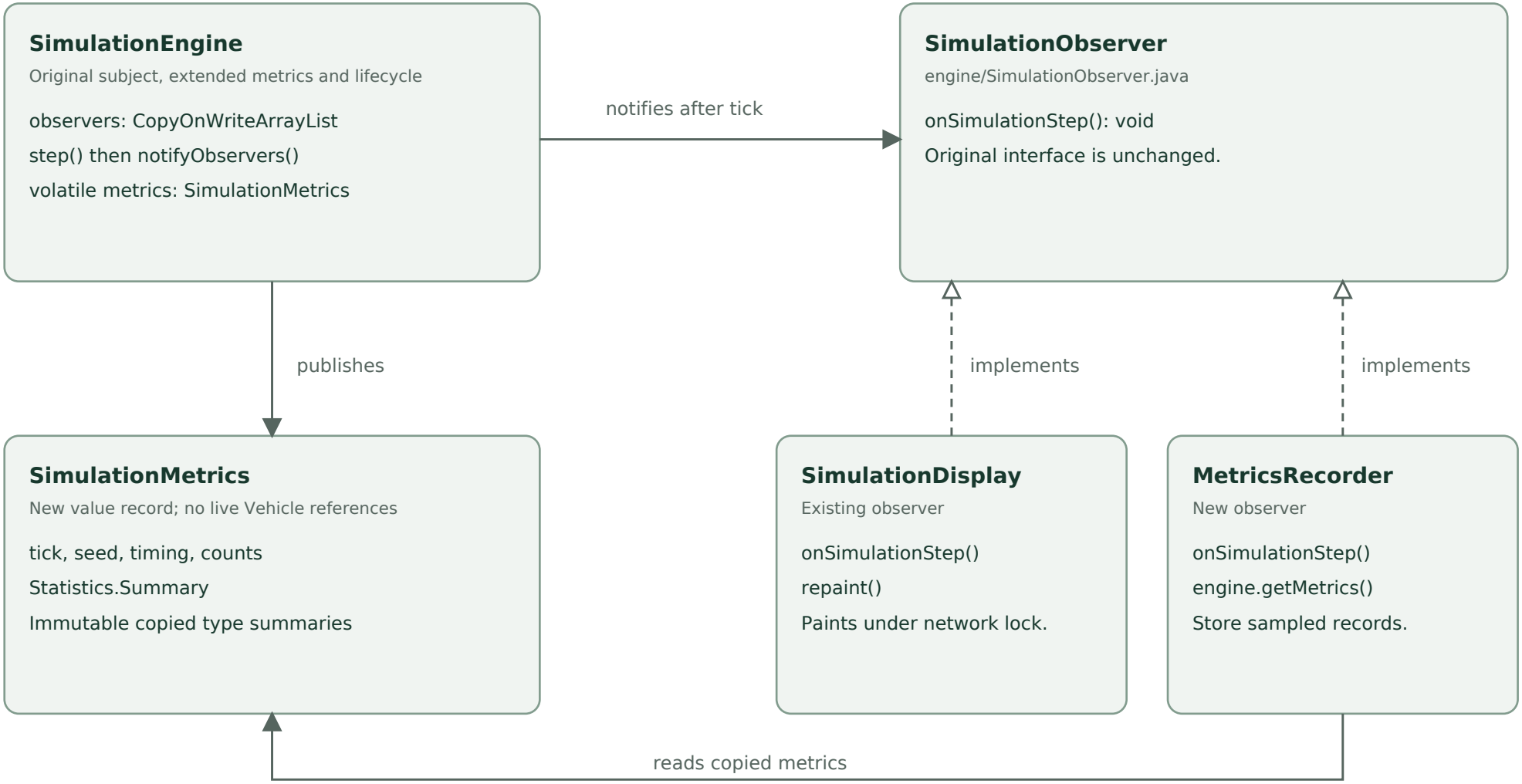
The original observer callback has no argument. The engine publishes completed metrics before notifying registered observers.



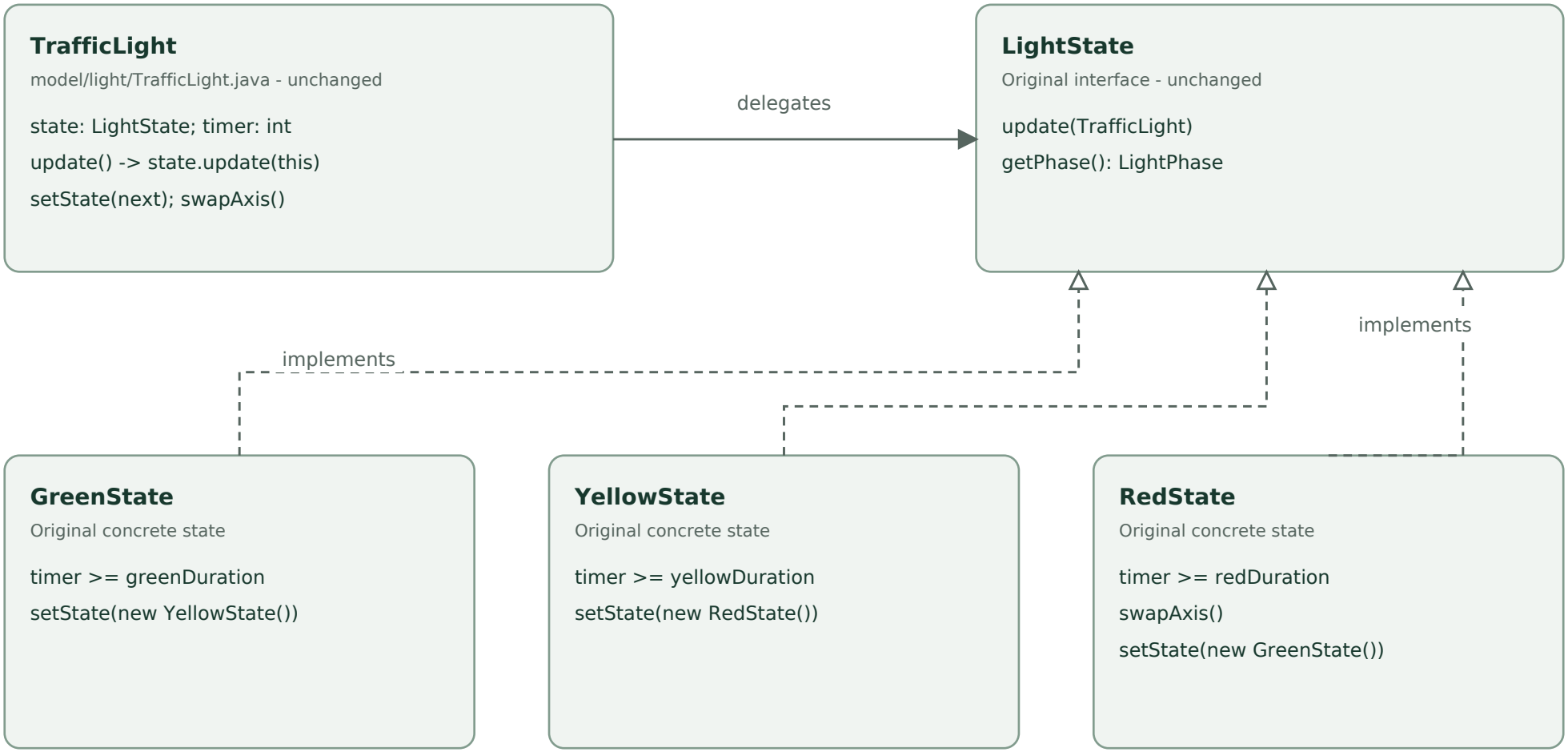
The model lock protects painting and mutation. MetricsRecorder reads an immutable volatile snapshot and synchronises its own sample list; CSV writing occurs outside the callback.

Names and interactions correspond to the supplied Java source. Arrows are labelled by relationship.

State | Retain the original signal transitions

COMP2000 | Rongrui Guo | 48977152 | Individual implementation

The three original LightState implementations are unchanged. The timer and active axis remain in TrafficLight.

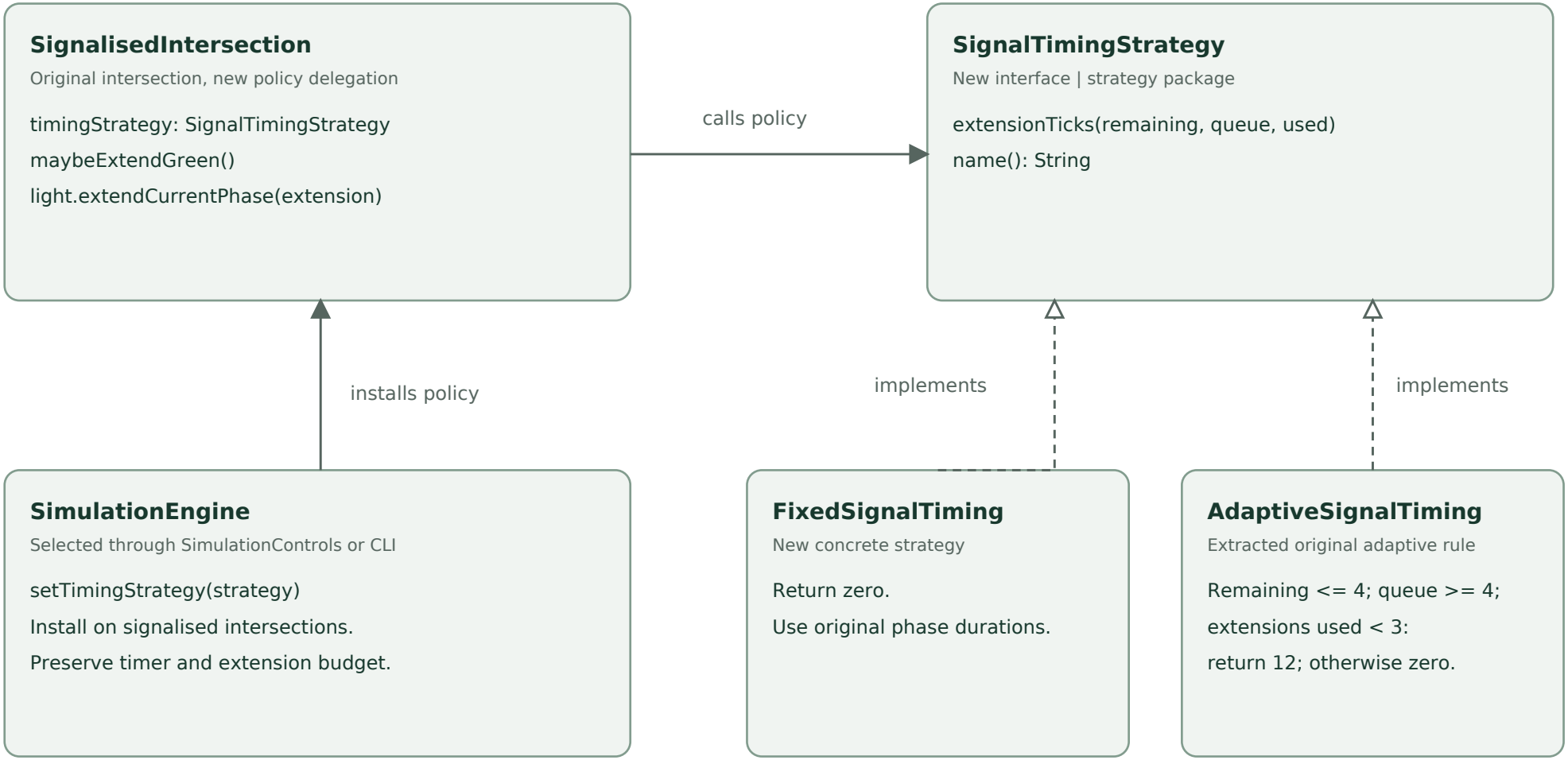


Scope: RedState uses a timer; it does not test occupancy. The original TrafficSensor separately prevents entry into conflicting occupied junctions and handles emergency yielding.

Strategy | Extract the original green-extension rule

COMP2000 | Rongrui Guo | 48977152 | Individual implementation

SignalisedIntersection is the context. Its default adaptive policy preserves the original rule; fixed timing adds no extension.



Regression evidence: the default adaptive version matches original vehicle and signal traces for three seeds over 3000 ticks each. Policy selection remains separate from the State pattern.